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Electronic apparatus capable of detecting unauthorized removal of screws from chassis

Abstract

A screw fixing portion of an electronic apparatus includes a screw, a screw hole formed in a bottom cover to insert the screw therethrough, a stud provided on a top cover to be threaded with the screw, a substrate hole through which the screw is inserted at a point interposed between the bottom cover and the stud at a main substrate, a conductive sponge interposed between the bottom cover and the main substrate and provided around the screw, and a contact pair which is arranged opposite to each other so as to surround the substrate hole on the surface of the main substrate and contacts and conducts with the conductive sponge by fastening the screw.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an electronic apparatus including an electronic substrate in a chassis in which a first cover and a second cover are fixed by screw fixing portions.

Description of the Related Art

(2) A structure in which two cover members are stacked and connected to be configured in a flat box shape has been widely used for a chassis of an electronic apparatus such as a notebook type personal computer (laptop PC) or a tablet type personal computer (tablet PC).

(3) In the laptop PC, there has been used as a main body chassis provided with a keyboard, a

structure body in which top and bottom two cover members are overlapped and both chassis members are screwed. Electronic parts such as a CPU, a memory, a battery, etc. have been housed in the internal space of the structure body.
(4) By the way, the screw can be easily removed with a screwdriver, and anyone can disassemble the cover members and operate the inside. Therefore, a tamper switch may be provided which detects that the cover members have been disassembled to prevent careless operation. When the tamper switch detects that the cover members have been removed, the input of a predetermined password is requested on the screen to enable the prevention of internal operations by unauthorized persons.
(5) [Patent Document 1] Japanese Patent No. 6986607
SUMMARY OF THE INVENTION
(6) The tamper switch as described above requires a certain degree of cost, and it is desirable to omit the tamper switch in order to further reduce the cost of the electronic apparatus. As a unit of omitting the tamper switch, it is only necessary to provide a predetermined electrical configuration to screw fixing portions fixing the two cover members to detect that the screw has been removed.
(7) However, even if a screw removal detection mechanism is provided in one screw fixing portion in a general laptop PC, two cover members can also be rotated relatively about the screw fixing portion provided with the detection mechanism if screws in other screw fixing portions are removed, thus resulting in exposure of the inside. Also, if there is only one screw detection mechanism, there is a concern of erroneous detection.
(8) The present invention has been made in view of the above-described problems. An object of the present invention is to provide an electronic apparatus capable of reliably detecting removal of two cover members.
(9) In order to solve the above-described problems and achieve the object, there is provided an electronic apparatus according to an aspect of the present invention, which has an electronic substrate included in a chassis in which a first cover and a second cover are fixed by a screw fixing portion. The screw fixing portion includes a screw, a screw hole formed in the first cover to insert the screw therethrough, a stud provided on the second cover to be threaded with the screw, a substrate hole through which the screw is inserted at a point interposed between the first cover and the stud at the electronic substrate, a conductor interposed between the first cover or the stud and the electronic substrate and provided around the screw, and a contact pair which is arranged opposite to each other so as to surround the substrate hole on the surface of the electronic substrate on the side facing the conductor and contacts and conducts with the conductor by fastening the screw. The screw fixing portion is provided in a plural number, and a plurality of the contact pairs form a series circuit in which the contact pairs are connected in series. One end of the series circuit is connected to one electrode of a power supply, and the other end of the series circuit is connected to the other electrode of the power supply via a pull resistor and connected to an input terminal of a controller chip.
(10) According to such an electronic apparatus, it is possible to collectively detect the states of the contact pairs in the plurality of screw fixing portions by a voltage applied to the input terminal of the controller chip. Further, since the screw fixing portion is provided in the plural number, even when the two cover members are rotated relatively about one screw fixing portion to expose the inside of the electronic apparatus, the contact pairs of other screw fixing portions become non-conductive and thereby the detection becomes possible. It is thus possible to reliably detect removal of the two cover members.
(11) Each of the screw fixing portions may have a voltage dividing resistor connected in parallel with the contact pair, and the input terminal may be a terminal capable of analog input. By providing the voltage dividing resistor in each contact pair in this way, it is possible to detect the number of points where the contact pairs become non-conductive.
(12) The controller chip may detect the number of points where the contact pair is made non-

conductive from a voltage applied to the input terminal, and perform operation restriction processing when the number of points is two or more. Thus, if only one contact pair is non-conductive, no operation restriction processing is performed, and erroneous detection and erroneous processing can be prevented.
(13) The resistance value of each of the voltage dividing resistors may differ from the total resistance value of any one or more of the other voltage dividing resistors excluding itself. According to such a configuration, it is possible to identify the point of the contact pair brought into non-conduction.
(14) The conductor may be a conductive sponge. The conductive sponge is suitable because it has conductivity and elasticity.
(15) The chassis may be rectangular, and the screw fixing portions may be provided at points along at least three edges of the chassis. Since the three edges are different in vibration mode, even if one contact pair becomes non-conductive due to the effect when some kind of vibration is applied, other points will not be affected by the vibration, and thereby erroneous detection can be prevented.
(16) According to the above-described aspect of present invention, since the screw fixing portion is provided in a plural number, it is possible to reliably detect removal of the first cover and the second cover.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of an electronic apparatus according to an embodiment of the present invention.
(2) FIG. **2** is an exploded perspective view of a main body chassis as viewed obliquely from below.
(3) FIG. **3** is a sectional side view of a screw fixing portion.
(4) FIG. **4** is a partial sectional schematic perspective view of the screw fixing portion.
(5) FIG. **5** is a circuit diagram illustrating a mode of connection between a contact pair and an embedded controller.
(6) FIG. **6** is a circuit diagram illustrating a mode of connection between the contact pair and the embedded controller in a state in which a bottom cover and a top cover are closed.
(7) FIG. **7** is a circuit diagram illustrating a mode of connection between the contact pair and the embedded controller in a state in which the bottom cover and the top cover are partially open therebetween.
(8) FIG. **8** is a circuit diagram according to a modification.
DETAILED DESCRIPTION OF THE INVENTION
(9) An embodiment of an electronic apparatus according to the present invention will hereinafter be described in detail with reference to the accompanying drawings. Incidentally, the present invention is not limited by this embodiment.
(10) FIG. **1** is a perspective view illustrating an electronic apparatus **10** which is an embodiment of the present invention. The electronic apparatus **10** is a laptop PC. The present invention can be used for various electronic apparatuses such as a desktop PC, a tablet PC, a smart phone or a mobile phone, etc. Further, the present invention can also be applied to a so-called foldable PC, which includes a foldable display (organic EL type, etc.), for example.
(11) As illustrated in FIG. **1**, the electronic apparatus **10** includes a main body chassis **14** having input units such as a keyboard device **16**, a touch pad **17**, etc., and a rectangular plate-shaped display chassis **20** having a display device **18** comprised of a liquid crystal display or the like.
(12) The main body chassis (chassis) **14** is a flattened rectangular box in which a bottom cover (first cover) **21** and a top cover (second cover) **22** are connected together. The bottom cover **21** and the top cover **22** are fixed by screw fixing portions **12** to constitute the main body chassis **14**. The

bottom cover **21** and the top cover **22** are, for example, a resin material. There are accommodated inside the main body chassis **14**, various electronic parts such as a main substrate (electronic substrate) **24**, a battery **26** (refer to FIG. **2**), etc. The keyboard device **16** and the touch pad **17** are arranged in the central portion of the top cover **22** which constitutes the upper surface of the main body chassis **14**.
(13) The display chassis **20** is connected to the trailing edge portion of the main body chassis **14** via hinges **23** so that it can be opened and closed. The display chassis **20** is electrically connected to the main body chassis **14** by an unillustrated cable having passed through each of the hinges **23**. The display device **18** is, for example, a liquid crystal display.
(14) FIG. **2** is an exploded perspective view of the main body chassis **14** as viewed obliquely from below. As illustrated in FIG. **2**, the main substrate **24**, the battery **26**, a fan **27**, and the like are accommodated in the top cover **22** in the main body chassis **14**. A CPU **24***a* and an embedded controller (controller chip) **24***b* are mounted on the main substrate **24**.
(15) The CPU **24***a* generally controls the entire electronic apparatus **10** based on an OS (Operating System). The embedded controller **24***b* performs basic processing based on a BIOS (Basic Input Output System) and a UEFI (Unified Extensible Firmware Interface). Power for the embedded controller **24***b* is a so-called real-time clock power supply and is also supplied from a predetermined backup battery in addition to the battery **26**, so that the embedded controller **24***b* can operate even when the electronic apparatus **10** is not activated or sleeps. The embedded controller **24***b* may be provided in a sub substrate different from the main substrate **24**.
(16) The screw fixing portions **12** are provided at a total of five points of a point along the trailing edge **14***a* of the main body chassis **14**, points along both side edges **14***b*, and each neighborhood of the corners of the trailing edge **14***a* and the side edges **14***b*. Assume that the two or more screw fixing portions **12** are provided.
(17) FIG. **3** is a sectional side view of the screw fixing portion **12**. FIG. **4** is a partially sectional schematic perspective view of the screw fixing portion **12**. The screw fixing portion **12** has a screw **30**, a screw hole **32**, a stud **34**, a substrate hole **36**, a conductive sponge (conductor) **38**, a contact pair **40**, and a voltage dividing resistor **42**.
(18) The screw hole **32** is formed in the bottom cover **21** and is a point through which the screw **30** is inserted. The screw hole **32** is formed in a bottom wall **44***a* of a recessed portion **44** in the bottom cover **21**. The stud **34** allows the screw **30** to be screwed therein and is fixed to an internal surface **22***a* in the top cover **22**. The substrate hole **36** is a point through which the screw **30** is inserted, and is formed at a point where it is interposed between the bottom wall **44***a* of the bottom cover **21** and the stud **34** at the main substrate **24**. The main substrate **24** includes a plurality of convexes **24***c* (refer to FIG. **2**), and some of the substrate holes **36** are formed in the convexes **24***c*.
(19) The conductive sponge **38** is a sponge having conductivity and elasticity. The conductive sponge **38** is interposed between the bottom wall **44***a* of the bottom cover **21** and the main substrate **24** and surrounds the periphery of a threaded portion **30***a* at the screw **30**. The conductive sponge **38** is annular and has a moderate thickness with which it is elastically sandwiched between the bottom wall **44***a* and the main substrate **24**. The conductive sponge **38** is fixed to the bottom wall **44***a* by an adhesive or an adhesive tape or the like. The conductor provided around the screw **30** between the bottom cover **21** and the main substrate **24** may be a metal washer, a metal foil, metal plating on the inner surface of the bottom wall **44***a*, etc. even in addition to the conductive sponge **38**.
(20) The contact pair is made up of two lands **40***a*. The two lands **40***a* respectively have a substantially circular arc shape with an angle slightly smaller than 180°. The two lands **40***a* are arranged opposite to each other via a narrow gap therebetween so as to surround the substrate hole **36** on the surface (lower surface **24***d* in the case of this embodiment) of the main substrate **24** on the side facing the conductive sponge **38**. The two lands **40***a* become conductive when the conductive sponges **38** come into contact therewith by the fastening of the screw **30**. The voltage

dividing resistor **42** is connected in parallel with the contact pair **40**. The voltage dividing resistor **42** is surface-mounted onto the main substrate **24**. A lower surface of the bottom cover **21** is covered with an insulating coating film **21***a*. In FIG. **4**, the contact pair **40** and its connecting lines are indicated by dots for easy recognition thereof.

(21) FIG. **5** is a circuit diagram illustrating a mode of connection between each contact pair **40** and the embedded controller **24***b*. The five contact pairs **40** form a series circuit **46** in which they are connected in series. One end of the series circuit **46** is connected to a ground **48** (one electrode of the power supply). The other end of the series circuit **46** is connected to a power supply **52** (the other electrode of the power supply) via a pull resistor **50** and connected to an input terminal **24***ba* of the embedded controller **24***b*. The input terminal **24***ba* is a terminal capable of analog input. The power supply **52** is the real-time clock power supply described above.

(22) In FIGS. **5** through **8**, codes R**1**, R**2**, R**3**, R**4**, and R**5** each indicating a resistance value are attached to the voltage dividing resistors **42** in the respective screw fixing portions **12**. Further, the resistance value of the pull resistor **50** is assumed to be R**0**. The resistance value of each of the voltage dividing resistors **42** is different from the total resistance value of any one or more of other voltage dividing resistors **42** excluding itself. For example, the resistance value R**1** is different from other resistance values R**2** to R**5** and is also different from the combined resistance value of two or more of the resistance values R**2** to R**5**.

(23) FIG. **6** is a circuit diagram illustrating a mode of connection between each contact pair **40** and the embedded controller **24***b* in the state in which the bottom cover **21** and the top cover **22** are closed. In FIGS. **6** and **7**, arrows indicate the flow of current. Thus, in the state in which the bottom cover **21** and the top cover **22** are fixed and closed by the five screw fixing portions **12**, each contact pair **40** is made conductive by the abutment of the conductive sponge **38** thereto, so that an input voltage detected by the embedded controller **24***b* becomes 0V. Since the voltage of the input terminal **24***ba* is 0V, the embedded controller **24***b* is capable of recognizing that the bottom cover **21** and the top cover **22** are in the state of being fixed and closed by the five screw fixing portions **12**.

(24) FIG. **7** is a circuit diagram illustrating a mode of connection between each contact pair **40** and the embedded controller **24***b* in the state in which the bottom cover **21** and the top cover **22** are partially open therebetween. Thus, when the screws **30** are removed at the screw fixing portions **12** corresponding to the voltage dividing resistors **42** of the resistance values R**2** and R**4**, and the conductive sponges **38** no longer exist, the contact pairs **40** in such parts become non-conductive. Then, as indicated by the arrow, the current flows through the voltage dividing resistors **42** so as to bypass at the points where the contact pairs **40** are non-conducting. Therefore, since a voltage divided by the resistance value R**0** and the resistance value (R**2**+R**4**) is applied to the input terminal **24***ba*, the embedded controller **24***b* recognizes that the bottom cover **21** and the top cover **22** are in the state in which they are partially open therebetween. In this case, when a prompt is made to enter a prescribed password on the screen under the action of the embedded controller **24***b*, and the correct password is not entered, predetermined operation restriction processing is performed to prevent operation by an unauthorized person.

(25) The embedded controller **24***b* can recognize based on the voltage value of the input at the input terminal **24***ba*, the number of points where the screws **30** and the conductive sponges **38** are removed from the five screw fixing portions **12**, and the contact pairs **40** are non-conducting. When it is recognized that the number of points is a predetermined number or more (for example, two or more), the embedded controller **24***b* performs the operation restriction processing via the above password input request. Consequently, although it is a rare case, even if the contact pair **40** becomes non-conductive only at one point due to some factors such as vibrations, it is possible to prevent erroneous detection and erroneous processing.

(26) Since the respective resistance values of the voltage dividing resistors **42** are different as described above, the embedded controller **24***b* can identify based on the input voltage of the input

terminal **24***ba*, at which portion of the five screw fixing portions **12**, the screw **30** and the conductive sponge **38** are removed. Further, since each resistance value is set to be different from the total resistance value of any one or more of other voltage dividing resistors **42** excluding itself, it is possible to distinguish and detect that the contact pair **40** at one predetermined point has become non-conductive, and the contact pairs **40** at other multiple points have become non-conductive. With this identification, for example, when the contact pair **40** at any point frequently becomes non-conductive, it is judged that a mechanical failure has occurred at this point. This is reflected in subsequent detection processing or can be used as reference data for ex-post design.
(27) Although there are provided the screw fixing portions **12** at the five points in the present embodiment, they are preferably provided in at least three points along the three edges at the main body chassis **14**, i.e., the trailing edge **14***a* and the side edges **14***b* on both sides. Since each edge is different in position and direction, the vibration mode differs. Even if the contact pair **40** at one point becomes non-conductive due to the effect of some kind of vibration applied thereto, other points are not affected by vibration, and the erroneous detection can be prevented.
(28) Even when a plurality of (N, for example) screw fixing portions **12** are provided, and the contact pairs **40** other than one of all the screw fixing portions **12** (N−1 points) become non-conductive, the embedded controller **24***b* naturally performs operation restriction processing. Therefore, even if the screw **30** is made slightly loose only at one point, the screws **30** of all other screw fixing portions **12** are removed, and the bottom cover **21** is rotated about the point where the screw **30** is made loose, to expose the inside of the main body chassis **14**, the embedded controller **24***b* can perform the operation restriction processing correctly based on the states of the four points where the screws **30** have been removed.
(29) Incidentally, depending on design conditions, in order just to cope with the fact that such a bottom cover **21** is rotated to expose the inside of the main body chassis **14**, the operation restriction processing may be performed when the contact pair **40** becomes non-conductive even at one of the plurality of screw fixing portions **12**. Since it is not necessary to identify the points where the contact pairs **40** become non-conductive or recognize the number of points, the voltage dividing resistor **42** in each screw fixing portion **12** may be omitted. Further, in this case, since the voltage applied to the input terminal **24***ba* of the embedded controller **24***b* becomes binary, a digital input terminal may be used.
(30) Thus, since the plurality of screw fixing portions **12** are provided in the electronic apparatus **10**, it is possible to reliably detect the removal of the bottom cover **21**. Further, since the contact pairs **40** of the screw fixing portions **12** are connected in series, the embedded controller **24***b* is capable of monitoring the state of one input terminal **24***ba*.
(31) FIG. **8** is a circuit diagram according to a modification. In the circuit diagram illustrated in FIG. **8**, the polarity is connected to be opposite to that illustrated in FIG. **5**. That is, one end of the series circuit **46** is connected to the power supply **52**, and the other end thereof is connected to the ground **48** via the pull resistor **50** and connected to the input terminal **24***ba*. Even in this circuit, only the polarities are different, and the same operation and effect as the above can be obtained.
(32) Incidentally, in the above embodiment, the contact pair **40** is provided in the lower surface **24***d* of the main substrate **24**, and the conductive sponge **38** is interposed between the main substrate **24** and the bottom wall **44***a* being a part of the first cover **21**, but conversely, as indicated by the virtual line of FIG. **3**, the contact pair **40** may be provided in the upper surface **24***e* of the main substrate **24**, and the conductive sponge **38** may be interposed between the main substrate **24** and the stud **34**. That is, the conductive sponge **38** may be interposed between the surface of the main substrate **24** in which the contact pair **40** is provided, and the first cover **21** or the stud **38**.
(33) It goes without saying that the present invention is not limited to above-described embodiments, and can be freely modified within the scope not departing from the gist of the present invention.

Claims

1. An electronic apparatus comprising: an electronic substrate in a chassis in which a first cover and a second cover are fixed by a plurality of screw fixing portions, wherein each screw fixing portion includes: a screw, a screw hole in the first cover and configured to receive the screw, a stud on the second cover and configured to thread with the screw, a substrate hole configured to receive the screw at a point between the first cover and the stud at the electronic substrate, a conductor between the first cover or the stud and the electronic substrate and configured to be disposed around the screw, and a contact pair arranged opposite to each other and configured to surround the substrate hole on a surface of the electronic substrate on a side facing the conductor, and is further configured to contact and conduct with the conductor by fastening the screw, wherein the plurality of screw fixing portions and a plurality of the contact pairs form a series circuit in which the contact pairs are connected in series, wherein one end of the series circuit is connected to one electrode of a power supply, and another end of the series circuit is connected to another electrode of the power supply via a pull resistor and connected to an input terminal of a controller chip; wherein each of the screw fixing portions has a voltage dividing resistor connected in parallel with the contact pairs, and wherein the input terminal is a terminal capable of analog input.
2. The electronic apparatus according to claim 1, wherein the controller chip detects a number of points where the contact pairs are made non-conductive, from a voltage applied to the input terminal, and performs operation restriction processing when the number of points is two or more.
3. The electronic apparatus according to claim 1, wherein the resistance value of each of the voltage dividing resistors is different from a total resistance value of any one or more of the other voltage dividing resistors.
4. The electronic apparatus according to claim 1, wherein the conductor is a conductive sponge.
5. The electronic apparatus according to claim 1, wherein the chassis is rectangular, and wherein the screw fixing portions are at points along at least three edges of the chassis.
6. An electronic apparatus comprising: an electronic substrate in a chassis in which a first cover and a second cover are fixed by a plurality of screw fixing portions, wherein each screw fixing portion includes: a screw, a screw hole in the first cover and configured to receive the screw, a stud on the second cover and configured to thread with the screw, a substrate hole configured to receive the screw at a point between the first cover and the stud at the electronic substrate, a conductor between the first cover or the stud and the electronic substrate and configured to be disposed around the screw, and a contact pair arranged opposite to each other and configured to surround the substrate hole on a surface of the electronic substrate on a side facing the conductor, and is further configured to contact and conduct with the conductor by fastening the screw, wherein the plurality of screw fixing portions and a plurality of the contact pairs form a series circuit in which the contact pairs are connected in series, wherein one end of the series circuit is connected to one electrode of a power supply, and another end of the series circuit is connected to another electrode of the power supply via a pull resistor and connected to an input terminal of a controller chip; and wherein each of the screw fixing portions has a voltage dividing resistor connected in parallel with the contact pairs.
7. An electronic apparatus comprising: an electronic substrate in a chassis in which a first cover and a second cover are fixed by a plurality of screw fixing portions, wherein each screw fixing portion includes: a screw, a screw hole in the first cover and configured to receive the screw, a stud on the second cover and configured to thread with the screw, a substrate hole configured to receive the screw at a point between the first cover and the stud at the electronic substrate, a conductor between the first cover or the stud and the electronic substrate and configured to be disposed around the screw, and a contact pair arranged opposite to each other and configured to surround the substrate hole on a surface of the electronic substrate on a side facing the conductor, and is further

configured to contact and conduct with the conductor by fastening the screw, wherein the plurality of screw fixing portions and a plurality of the contact pairs form a series circuit in which the contact pairs are connected in series, wherein one end of the series circuit is connected to one electrode of a power supply, and another end of the series circuit is connected to another electrode of the power supply via a pull resistor and connected to an input terminal of a controller chip; and wherein the input terminal is a terminal capable of analog input.